Bayesian Updating

UBCO MDS — DATA 582

Bayesian vs Frequentist

- Bayesians consider the path to inference by assuming variability in the parameters θ. Notably, they consider the observed data $\mathbf{x}$ FIXED.
- Frequentists consider the path to inference by assuming variability in the observed data $\mathbf{x}$. Notably, they consider the parameters θ FIXED.
- For Bayesian, we are seeking $h(\theta \mid \mathbf{x})$: the posterior distribution of θ.

Bayesian

- Recall the simplest form for us to consider:

$$h(\theta \mid \mathbf{x}) \propto f(\mathbf{x} \mid \theta) g(\theta)$$

- The posterior distribution of θ depends on the observed data (through the likelihood) and the prior distribution we imposed on θ.

- One way to think of this is that we start with a prior distribution for θ and then update that distribution when we observe some data.

- Today, we get into some details and examples.

Beta-Bernoulli Example

- Suppose $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$, aka has pmf

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

- We wish to perform inference on p, using the Bayesian approach.
- What do we need to do next?

Beta-Bernoulli Example

- We will suggest the Beta distribution as a sensible parent distribution for p, that would be flexible enough to capture whatever prior knowledge we might have about p...

[1]

[1] By Horas based on the work of Krishnavedala - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=15404515

Beta-Bernoulli Example

- So, we have the following framework
- Likelihood function from n Bernoulli trials:

$$\begin{aligned} f(\mathbf{x} \mid p) &= \left(p^{x_1}(1-p)^{x_1}\right)\left(p^{x_2}(1-p)^{x_2}\right)\cdots\left(p^{x_n}(1-p)^{x_n}\right) \\ &= p^{\sum x_i}(1-p)^{(n-\sum x_i)} \end{aligned}$$

- Prior distribution for p:

$$g(p) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}$$

(note: $\alpha, \beta > 0$)

Beta-Bernoulli Example

- Reminder: $h(p \mid \mathbf{x}) \propto f(\mathbf{x} \mid p)g(p)$
- So we multiply:

$$h(\theta \mid \mathbf{x}) \propto p^{\sum x_i}(1-p)^{(n-\sum x_i)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}$$

Beta-Bernoulli Example

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- So we multiply:

$$h(p \mid \mathbf{x}) \propto p^{\sum x_i}(1-p)^{(n-\sum x_i)} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}$$
$$\propto \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\sum x_i+\alpha-1}(1-p)^{(n-\sum x_i)+\beta-1}$$

- Well, now what?

Beta-Bernoulli Example

- We've figured out that our posterior distribution for p is proportional to

$$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\sum x_i+\alpha-1}(1-p)^{(n-\sum x_i)+\beta-1}$$

- Remember, p is the random variable here, so anything that is not a function of p is some sort of constant...

$$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\sum x_i+\alpha-1}(1-p)^{(n-\sum x_i)+\beta-1}$$

Beta-Bernoulli Example

- So then, our posterior distribution for p is also proportional to

$$p^A(1-p)^B$$

- Our prior had assumed p was continuous and bounded $[0,1]$. Is there a known continuous distribution with these assumptions and this general form for the random variable?
- Hint...we've already been using it this lecture...

Beta-Bernoulli Example

- The Beta distribution has the form, for a random variable p

$$\frac{\Gamma(A+B)}{\Gamma(A)\Gamma(B)} p^{A-1}(1-p)^{B-1}$$

(note: $A, B > 0$)

- So, with a slight shift on the constants from $p^A(1-p)^B \to p^{A-1}(1-p)^{B-1}$ I can see that

$$h(p \mid \mathbf{x}) \propto p^{A-1}(1-p)^{B-1} \propto \frac{\Gamma(A+B)}{\Gamma(A)\Gamma(B)} p^{A-1}(1-p)^{B-1}$$

- Since the right-hand side *is* a probability distribution that integrates to 1, we now know that $h(p \mid \mathbf{x})$ is Beta(A, B).

Beta-Bernoulli Example

- We can take this to further detail though...we can solve for A and B!
- Remember our initial proportional form for $h(p \mid \mathbf{x})$ had constants

$$
\begin{aligned}
h(p \mid \mathbf{x}) &\propto \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\sum x_i+\alpha-1}(1-p)^{(n-\sum x_i)+\beta-1} \\
&\propto p^{\sum x_i+\alpha-1}(1-p)^{(n-\sum x_i)+\beta-1} \\
&= p^{A-1}(1-p)^{B-1}
\end{aligned}
$$

- And so we know that $A = \sum x_i + \alpha$ and $B = (n - \sum x_i) + \beta$

Beta-Bernoulli Example

- Bringing it all together. If pre-assume
  - $X \sim \text{Bernoulli}(p)$
  - $p \sim \text{Beta}(\alpha, \beta)$
- then
  - $p \mid \mathbf{x} \sim \text{Beta}(\alpha + \sum x_i, \beta + n - \sum x_i)$

- We can therefore see, mathematically, how our observed data (through the total number of Bernoulli successes in n trials) updates our prior distribution of p to our posterior distribution of p.

- As such, our inferences will be affected by both our prior belief surrounding p, as well as our observed data through $\mathbf{x}$.

Some Comments

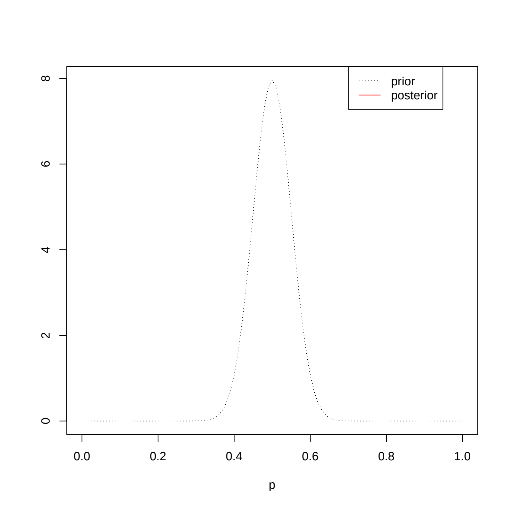
- The fact that our prior and posterior ended up being from the same family (Beta, here) is *definitely* not guaranteed.
- This phenomenon has its own terminology: conjugate prior.
- The biggest benefit we get from conjugate priors is that the posterior is in closed-form, which allows for simple updating. So, they are often sought after, when possible.
- When we don't get closed-form posterior distributions, we will have to develop some other computational tricks to sort things out.
- Let's actually deal with some data...

Beta-Bernoulli: Coin Spin

- Let $X_1, X_2, \ldots, X_n$ represent the shown side of a coin, and let tails be considered a success (1) and heads be considered a failure (0).
- In intro stats, we generally set up an experiment like this with a 'fair coin flip'.
- Spinning, rather than flipping, a coin is more sensitive to weight imbalances (and other imperfections) that might bias the end result. How much? I don't really know...
- Suppose we pick up a well-used loonie and intend to spin it for this experiment.

Beta-Bernoulli: Coin Spin

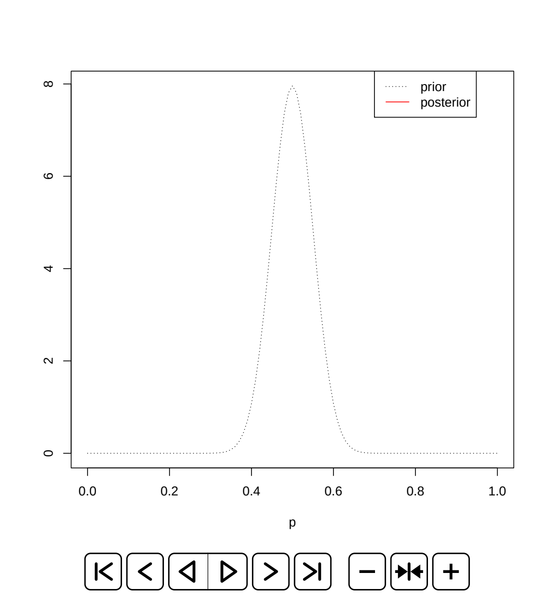
- We decide that Beta($\alpha = 50, \beta = 50$) seems like a sensible prior distribution for p (probability of tails showing)...



Beta-Bernoulli: Coin Spin

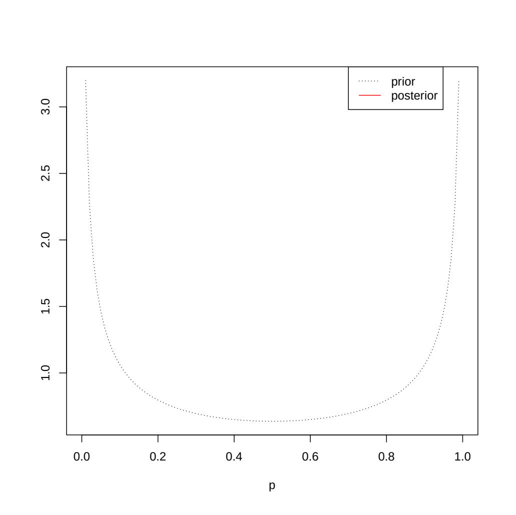
- Now suppose the well-worn loonie, against all odds, actually was perfectly weighted such that the probability was 0.5.
- We start spinning and recording the shown face...

Bayesian Updating - Good Prior

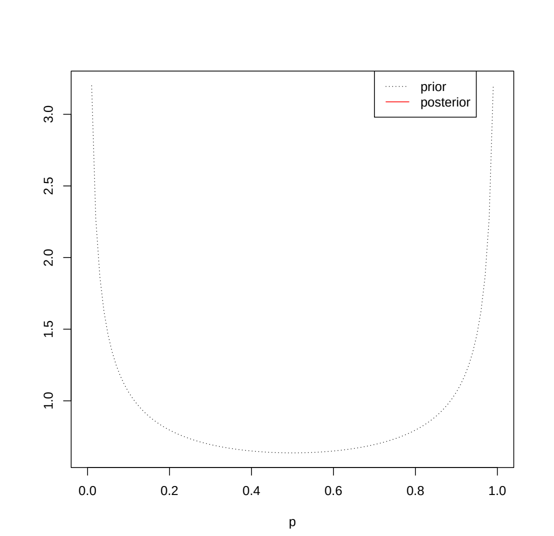


Beta-Bernoulli: Coin Spin

- What if we had suspected that it was a massively weighted coin, but we didn't know towards which face: Beta(0.5, 0.5)

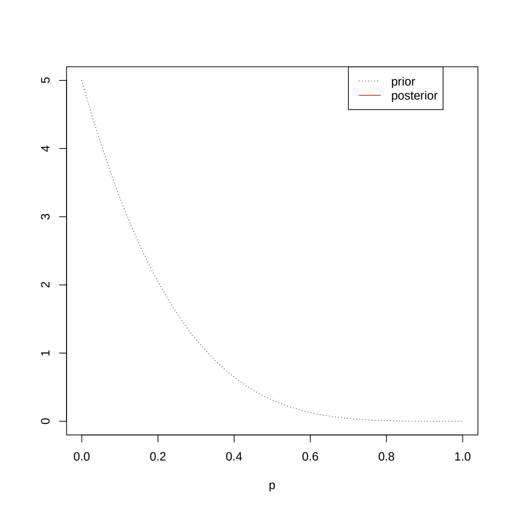


Bayesian Updating - Extremes Prior

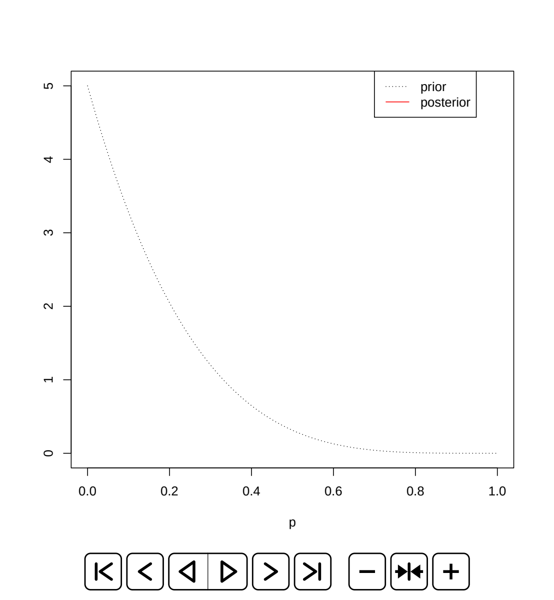


Beta-Bernoulli: Coin Spin

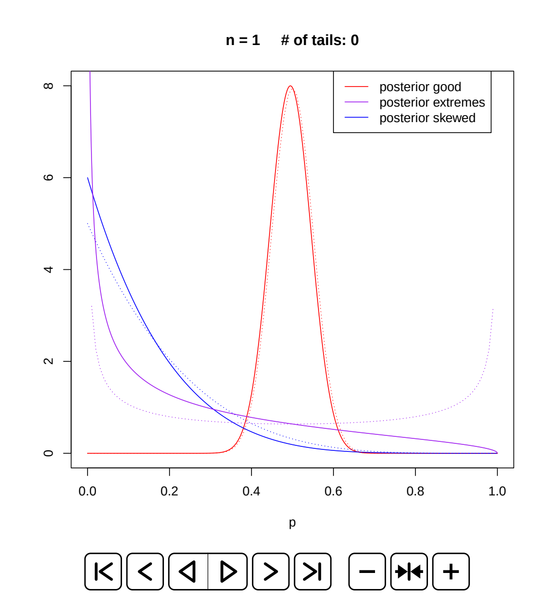
- What if we had suspected that it was a weighted towards heads: Beta(1, 5)



Bayesian Updating - Skewed Prior



Bayesian Updating - All Overlayed

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